MATH2921 Year 2023 Quiz 2. ①

Solution Sheet.

Q1. Solution :

i) $z = f(x,y) = -x^2+y^2$

$(-f_x, -f_y, 1)$

$= (2x, -2y, 1)$ ← upward

$$\text{Flux} = \iint_S \vec{F}\cdot\vec{n}\, dS$$ ← unit upward normal

$$\left(\vec{n} = \frac{(2x,-2y,1)}{\sqrt{4x^2+4y^2+1}}\right)$$

$R = \{(x,y) \mid x^2+y^2 \le 4\} \subset \mathbb{R}^2$

$z=f(x,y)=-x^2+y^2$

$$\text{Flux} = \iint_R (y, x, -(-x^2+y^2)) \cdot (2x,-2y,1)\, dx\,dy$$

$$= \boxed{\iint_R (x^2-y^2)\, dx\,dy}$$

②

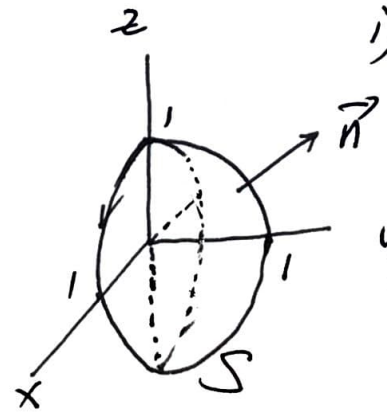
ii) $\iint_R (x^2-y^2)\, dx\,dy$

$$= \iint_R (r^2\cos^2\theta - r^2\sin^2\theta)\, r\, dr\, d\theta$$

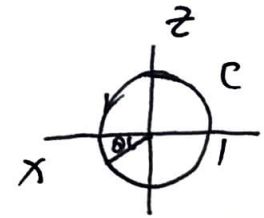
$$= \int_0^{2\pi}\int_0^2 r^3\cos 2\theta\, dr\, d\theta$$

$$= \underbrace{\int_0^{2\pi}\cos 2\theta\, d\theta}_{0} \cdot \int_0^2 r^3\, dr = \boxed{0}.$$

Q2. Solution :



i)



$\theta \in [0, 2\pi)$

$$\boxed{C: \vec{r}(\theta) = (\cos\theta, \overset{y}{0}, -\sin\theta)}$$

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\, dS = \oint_C \vec{F}\cdot d\vec{r}$$

$$= \boxed{\oint_C (0, 0, \sin^2\theta\cos\theta)\cdot d\vec{r}}$$

ii) $$= \int_0^{2\pi} (0, 0, \sin^2\theta\cos\theta) \cdot (-\sin\theta, 0, -\cos\theta)\, d\theta$$

$$= \int_0^{2\pi} -\sin^2\theta\cos^2\theta \, d\theta$$

$$= -\frac{1}{4}\int_0^{2\pi} \sin^2 2\theta \, d\theta$$

$$= -\frac{1}{4}\int_0^{2\pi} \frac{1-\cos 4\theta}{2} \, d\theta$$

$$= -\frac{1}{8}\int_0^{2\pi} d\theta = -\frac{\pi}{4}.$$

Q3. Solution:

i) $e^A \cdot e^{-A} = e^{A-A} = e^0 = I$

$\boxed{\text{True}}$ $(e^A)^{-1} = e^{-A}$.

$(A \cdot (-A) = (-A) \cdot A = -A^2)$

[Tutorial Sheet 7, Question 4]

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\det(\lambda I_2 - A) = \begin{vmatrix} \lambda-1 & -2 \\ -2 & \lambda-1 \end{vmatrix}$$

$$= (\lambda-1)^2 - 4 = 0.$$

$\lambda - 1 = \pm 2$. $\lambda = -1$ or 3.

$\lambda = -1$: $\begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$\lambda = 3$: $\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

So $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1}$

$$e^A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{-1} & 0 \\ 0 & e^3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{-1} & 0 \\ 0 & e^3 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{-1} & e^3 \\ -e^{-1} & e^3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= \boxed{\frac{1}{2} \begin{pmatrix} e^{-1}+e^3 & -e^{-1}+e^3 \\ -e^{-1}+e^3 & e^{-1}+e^3 \end{pmatrix}}.$$

Q4. Solution: i)

$$y'' + \frac{1}{x} y' - \frac{10}{x^2} y = 0$$

$$p(x) = 1/x.$$

$$W(x)' = -p(x)\, W(x)$$

$$\boxed{W' = -\frac{1}{x} W}.$$

ii)

$$W(2) = W(1)\, e^{\int_1^2 -\frac{1}{x}\,dx}$$

$$= 1 \cdot e^{-\ln x \big|_1^2}$$

$$= e^{-\ln 2}$$

$$= \boxed{1/2}$$

Q5. Solution: i)

$$y'' - 2y' - 3y = 0. \qquad \lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0. \qquad \lambda = 3, -1.$$

Try $y_p(x) = \boxed{A + Bxe^{-x}.}$ $\quad A \rightsquigarrow 1$, $Bxe^{-x} \rightsquigarrow e^{-x}$

ii) $y_p = A + Bxe^{-x}$

$y_p' = Be^{-x} - Bxe^{-x}$

$y_p'' = -Be^{-x} - Be^{-x} + Bxe^{-x}$

$= -2Be^{-x} + Bxe^{-x}$

Plug into the ODE:

Constant: $-3A = 1 \rightsquigarrow A = -\frac{1}{3}$

e^{-x}: $-2B - 2(B) = 1 \rightsquigarrow B = -\frac{1}{4}$

xe^{-x}: $B - 2(-B) - 3B = 0$ ✓.

$$\boxed{y_p(x) = -\frac{1}{3} - \frac{1}{4}xe^{-x} \quad (\text{or with } c_1e^{3x} + c_2e^{-x})}$$